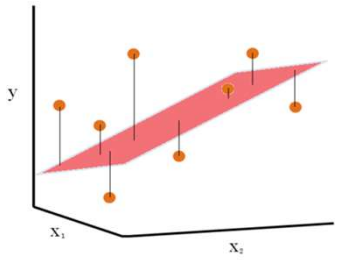
Agenda

- Earlier
  - Distribution of the estimators
  - Unknown variance
- Today
  - Multiple Linear regression



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Least-squares optimization

- Simple linear model: $Y = \alpha + \beta x + \epsilon$, ϵ a random variable $\mathbb{E}[\epsilon] = 0$
  - Training data: $\{(x_i, y_i): i = 1 \ldots n\}$
  - α and β unknown, estimated using estimators A and B
- Want to minimize $\sum_{i=1}^{n}(Y_i - Bx_i - A)^2$ which is done either via calculus or linear algebra
- Result in the normal equations (estimator form)
- Observed-data form of solution
- One can show the estimators are *unbiased*

$$\sum_{i=1}^{n} x_i Y_i = B\sum_{i=1}^{n} x_i^2 + A\sum_{i=1}^{n} x_i,$$

$$\sum_{i=1}^{n} Y_i = B\sum_{i=1}^{n} x_i + nA$$

$$b_1 := b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(\sum_{i=1}^{n} x_i y_i) - n\bar{x}\bar{y}}{(\sum_{i=1}^{n} x_i^2) - n\bar{x}^2} \left(= r\frac{s_Y}{s_X}\right) \qquad b_0 := a = \bar{y} - b_1\bar{x}$$

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Least-squares via normality assumption

- To find how good the estimator is we make additional assumptions on the error
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - Likelihood: $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta\, x_i - \alpha)^2}{2\sigma^2}\right]$
  - Log likelihood $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
- We reach the same equation as before for estimating α and β
- With $S_{xx} = \sum_i \mathrm{x}_i^2 - n\bar{\mathrm{x}}^2$, the distributions are obtained:
  - $B \sim \mathcal{N}(\beta, \frac{\sigma^2}{S_{xx}})$ and $A \sim \mathcal{N}(\alpha, \frac{\sigma^2 \sum_i \mathrm{x}_i^2}{nS_{xx}})$

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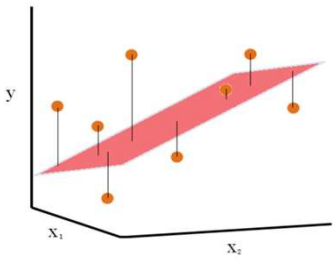
Variance

- Define $\mathrm{SSR} = \sum_{i=1}^{n}(Y_i - A - B\mathrm{x}_i)^2$
- With $S^2 = \frac{\mathrm{SSR}}{n-2}$, we have $\mathbb{E}[S^2] = \sigma^2$ (unbiased) (requires uncorrelated ϵ_i)
- With additional assumption of normality
  - biased MLE estimator for σ^2: $\boxed{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathrm{B}x_i - \mathrm{A})^2}$
  - Distribution $\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_{n-2}$

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Multiple Linear Regression

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \dots x_{ik}, y_i): i = 1 \dots n\}$
  - $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
  - $B=(B_0,\dots,B_k)^T$ estimates $\beta=(\beta_0,\dots,\beta_k)^T$



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Multiple Linear Regression via normality

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \dots x_{ik}, y_i): i = 1 \dots n\}$
  - $Y \mid x_1, \dots, x_k \sim \mathcal{N}(\mu_x, \sigma^2)$
    - $\mu_x = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$
- $L(\beta_0, \dots, \beta_k, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2}{2\sigma^2}\right]$
- $\ell = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2 - \frac{n}{2} \log(2\pi\sigma^2)$
- Applying the maximizing rule to random responses gives the estimator $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$

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Normal Equations

$$\ell(\beta_0, \ldots, \beta_k, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right)^2 - \frac{n}{2} \log(2\pi\sigma^2)$$

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0$$

$$\frac{\partial \ell}{\partial \beta_1} = \frac{1}{\sigma^2} \sum_{i=1}^{n} x_{i1} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0$$

$$\frac{\partial \ell}{\partial \beta_r} = \frac{1}{\sigma^2} \sum_{i=1}^{n} x_{ir} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right) = 0, \quad r = 1, \ldots, k$$

Replacing the observed responses y_i by the random responses Y_i makes the minimizing coefficients the estimators B_0 … B_k (next slide)

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Normal Equations

$$\sum_{i=1}^{n} Y_i = B \sum_{i=1}^{n} x_i + nA$$

$$\sum_{i=1}^{n} x_i Y_i = B \sum_{i=1}^{n} x_i^2 + A \sum_{i=1}^{n} x_i$$

Earlier simple linear regression

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1 \sum_{i=1}^{n} x_{i1} + B_2 \sum_{i=1}^{n} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}$$

$$\sum_{i=1}^{n} x_{i1} Y_i = B_0 \sum_{i=1}^{n} x_{i1} + B_1 \sum_{i=1}^{n} x_{i1}^2 + B_2 \sum_{i=1}^{n} x_{i1} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{i1} x_{ik}$$

$$\sum_{i=1}^{n} x_{ik} Y_i = B_0 \sum_{i=1}^{n} x_{ik} + B_1 \sum_{i=1}^{n} x_{ik} x_{i1} + B_2 \sum_{i=1}^{n} x_{ik} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}^2$$

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Multiple Linear Regression

- $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
  - Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
- Rewrite all the n equations in matrix form

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \qquad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \qquad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \qquad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$Y \in \mathbb{R}^{n\times 1}, \quad X \in \mathbb{R}^{n\times p}, \quad \beta \in \mathbb{R}^{p\times 1}, \quad \epsilon \in \mathbb{R}^{n\times 1} \qquad p = k + 1$$

$$Y = X\beta + \epsilon$$

This is now a matrix notation

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Assemble Matrices

$$X^T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1k} & x_{2k} & \cdots & x_{nk} \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \quad Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X^TY = \begin{bmatrix} \sum_i Y_i \\ \sum_i x_{i1}Y_i \\ \vdots \\ \sum_i x_{ik}Y_i \end{bmatrix}$$

$$X^TX = \begin{bmatrix} n & \sum_i x_{i1} & \sum_i x_{i2} & \cdots & \sum_i x_{ik} \\ \sum_i x_{i1} & \sum_i x_{i1}^2 & \sum_i x_{i1}x_{i2} & \cdots & \sum_i x_{i1}x_{ik} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_i x_{ik} & \sum_i x_{ik}x_{i1} & \sum_i x_{ik}x_{i2} & \cdots & \sum_i x_{ik}^2 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} B_0 \\ B_1 \\ \vdots \\ B_k \end{bmatrix}$$

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Normal Equations

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1 \sum_{i=1}^{n} x_{i1} + B_2 \sum_{i=1}^{n} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}$$

$$\sum_{i=1}^{n} x_{i1} Y_i = B_0 \sum_{i=1}^{n} x_{i1} + B_1 \sum_{i=1}^{n} x_{i1}^2 + B_2 \sum_{i=1}^{n} x_{i1} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{i1} x_{ik}$$

$$\sum_{i=1}^{n} x_{ir} Y_i = B_0 \sum_{i=1}^{n} x_{ir} + \sum_{j=1}^{k} B_j \sum_{i=1}^{n} x_{ir} x_{ij}, \quad r = 1, \ldots, k$$

$$\sum_{i=1}^{n} x_{ik} Y_i = B_0 \sum_{i=1}^{n} x_{ik} + B_1 \sum_{i=1}^{n} x_{ik} x_{i1} + B_2 \sum_{i=1}^{n} x_{ik} x_{i2} + \cdots + B_k \sum_{i=1}^{n} x_{ik}^2$$

$$X^T Y = X^T X B$$

$$B = (X^T X)^{-1} X^T Y$$

Assumes invertibility

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Example

- Diameter of a tree is measured at a standardized point and is a function of the age, elevation, rainfall and specific gravity
- We can, for instance, ignore the specific gravity and compute the estimated linear regression model to be $y = 11.54873 + 0.05728x_1 + 0.08712x_2 + 7.33231x_3$



	Age (years)	Elevation (1,000 ft)	Rainfall (inches)	Specific Gravity	
1	44	1.3	250	.63	18.1
2	33	2.2	115	.59	19.6
3	33	2.2	75	.56	16.6
4	32	2.6	85	.55	16.4
5	34	2.0	100	.54	16.9
6	31	1.8	75	.59	17.0
7	33	2.2	85	.56	20.0
8	30	3.6	75	.46	16.6
9	34	1.6	225	.63	16.2
10	34	1.5	250	.60	18.5
11	33	2.2	255	.63	18.7
12	36	1.7	175	.58	19.4
13	33	2.2	75	.55	17.6
14	34	1.3	85	.57	18.3
15	37	2.6	90	.62	18.8

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Estimators are unbiased

$$
\begin{aligned}
E[\mathbf{B}] &= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}] \\
&= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta}+\mathbf{e})] \quad \text{since } \mathbf{Y}=\mathbf{X}\boldsymbol{\beta}+\mathbf{e} \\
&= E[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e}] \\
&= E[\boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e}] \\
&= \boldsymbol{\beta}+(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top E[\mathbf{e}] \\
&= \boldsymbol{\beta}
\end{aligned}
$$

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Variance

- Define SSR as before as the sum of the squares of the residuals
  - $\Sigma_i(Y_i - B_o - \Sigma_j B_j x_{ij})^2$
- With $S^2 = \frac{\text{SSR}}{n-\text{k}-1}$, we have $\mathbb{E}[S^2] = \sigma^2$ (unbiased)

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Multivariate Normal Distributions

- The least square estimators $B_0, B_1, \ldots B_k$ which are the elements of the matrix B are linear combinations of the independent normal random variables Y_i
- The set of independent random variables form an interesting object
  - The multivariate normal distribution
- We say the estimator vector B has a multivariate normal distribution

$$B = (X^T X)^{-1} X^T Y$$

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